

Otavio Paz Nascimento

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PROFESSIONAL SUMMARY

Electrical Engineering student at Columbia University with research experience in reinforcement learning, humanoid robotics, and soft robotic systems. Built and deployed a Sim2Real locomotion pipeline for the Unitree G1 using Isaac Lab and RSL-RL PPO; led independent soft robotics research under a competitive college fellowship. Demonstrated ability to work across the full research cycle: design, prototyping, testing, documentation, and dissemination. Seeking research roles in robotics, embodied AI, or related areas.

EDUCATION

Columbia University

B.S. in Electrical Engineering. Columbia Robotics: Webmaster, RoboCONN Committee, Humanoid Robot Team Member; Brazilian Society: Member.

New York, NY

May 2027

Oberlin College

B.A. in 3-2 Engineering. Robotics Club: President; Brazilian Association: Secretary; International Student Org.: Program Coordinator.

Oberlin, OH

May 2025

EXPERIENCE

Student Researcher (Prof. Dr. Yiannis Kantaros)

Kantaros Lab, WashU in St. Louis

June 2025 – August 2025

St. Louis, MO

- Built a reinforcement learning pipeline for the Unitree G1 humanoid robot using Isaac Lab 4.5.0.
- Trained a Sim2Real locomotion policy via RSL-RL PPO, achieving stable walking in under 20 hours of training.
- Set up a humanoid robotics workstation: dual-boot Linux, dependency management, and deployment safety procedures.
- Produced thorough technical documentation covering failures, deployment steps, and debugging procedures.

Lead Energy Detective

Office of Energy and Sustainability, Oberlin College

October 2022 – May 2025

Oberlin, OH

- Analyzed campus energy and water usage via BuildingOS, managing 10+ spreadsheets across 80+ buildings.
- Identified 20+ malfunctioning meters, uncovering a loss of \$30,000+ for the city.
- Built and documented data acquisition validation software in Python Tkinter, finding 130+ inconsistencies from 2022 to 2025.

PROJECTS

Design and Fabrication of Soft Robotic Actuators using Dragon Skin and Iron Powder

June 2023 – July 2023

Student Researcher, Oberlin College Research Fellowship

- Prototyped muscle-like actuators using Dragon Skin infused with iron powder, evaluating 5+ designs through 25+ iterations.
- Reduced total fabrication cost to under \$900 against a \$2,000 budget, a 54% reduction through low-cost soft robotics methods.
- Applied techniques in FDM 3D printing, silicone molding, and data analysis throughout design and testing.

Renewable Energy Hybrid System at Oberlin College

September 2023 – October 2025

Co-Developer

- Co-designed a 1.7kW off-grid solar panel and micro wind turbine system for Shagbark Learning Center.
- Secured \$3,500 grant from REpowering Schools and \$1,600 from Green Edge Fund to support project implementation.
- Presented at RE+ 2023 in Las Vegas and DWEA 2024, earning industry recognition.

Papa Tick (IFSP/Salto and Biological Institute of Sao Paulo)

October 2019 – January 2022

Co-Inventor and Product Manager

- Developed and patented a solar-powered electric trap system, extending tick attraction duration by 80%.
- Implemented firmware functionality in C, validating system performance through rigorous lab testing.
- Designed enclosure in SolidWorks 3D CAD and produced circuit diagrams for the embedded system.

SKILLS

Robotics and Simulation: IsaacLab, MuJoCo, ROS, Gazebo, RL, SolidWorks, Arduino

Programming: Python, C/C++, Rust, MATLAB, Docker

Fabrication: 3D Printing (FDM), Silicone Molding

Tools: Adobe Photoshop, After Effects, Premiere, Microsoft Office

Languages: Portuguese (Native), English (Advanced), Spanish (Intermediate), Brazilian Sign Language (Intermediate), American Sign Language (Basic)